

The Config Bench

V8 — Ohm's Law: Questions & Giveaway

Six Questions | Answer Key | On-Camera Script | @TheConfigBench | June 2026

Giveaway Rules: Subscribe to @TheConfigBench + Answer ALL questions correctly in the comments + **Show your math** — no work shown = no entry. Prize: FNIRSI DSO152 Mini Oscilloscope. Winner announced in V9.

V8 — Opening Script (approx. 45 sec)

CAMERA: Canon EOS 60D wide bench shot. All three breadboards visible or stacked in frame. Both meters. Components. Acer PM1610 in background. This is the payoff episode — dress the bench fully.

JOHN: Welcome back to The Config Bench. I'm John — and this is V8, the last video in the Ohm's Law series.

JOHN: We've covered Lab 1, Lab 2, and Lab 3. Single resistors, series circuits, parallel circuits. You've seen the math, you've seen the meters confirm it.

JOHN: Now it's your turn.

CAMERA: Look directly into camera.

JOHN: I've got six questions. All Ohm's Law. All different values than what we used in the labs — so you have to do the math yourself. Drop your answers in the comments with your work shown. Every step. If you just drop a number, it doesn't count.

JOHN: One subscriber who answers all six correctly and shows their work wins the FNIRSI DSO152 mini oscilloscope. I'll announce the winner in V9.

JOHN: Let's go through them together.

The Six Questions — On-Camera Walkthrough

Q1 — Single Resistor — Find the Current

You have a 120Ω resistor connected to a 6V battery. What is the current flowing through the circuit?

Given: $V = 6V$, $R = 120\Omega$

Find: Current (I)

✓ *Formula:* $I = V \div R$

✓ $I = 6 \div 120 = 0.05A$

✓ *Answer:* 50mA

Q2 — Series Circuit — Find Current & Voltage Drop

Two resistors are wired in series: $R_1 = 100\Omega$, $R_2 = 150\Omega$. Supply voltage is 12V. What is the total resistance, the current, and the voltage drop across each resistor?

Given: $V = 12V$, $R_1 = 100\Omega$, $R_2 = 150\Omega$

Find: R_{total} , I , V_{R1} , V_{R2}

✓ $R_{total} = R_1 + R_2 = 100 + 150 = 250\Omega$

✓ $I = V \div R_{total} = 12 \div 250 = 0.048A = 48mA$

✓ $V_{R1} = I \times R_1 = 0.048 \times 100 = 4.8V$

✓ $V_{R2} = I \times R_2 = 0.048 \times 150 = 7.2V$

✓ *Check:* $V_{R1} + V_{R2} = 4.8 + 7.2 = 12V$ ✓

Q3 — Series Circuit — Find the Missing Resistance

A series circuit has a 9V supply. The current measured is 30mA. One resistor is 100Ω . What is the value of the second resistor?

Given: $V = 9V$, $I = 30mA$ (0.03A), $R_1 = 100\Omega$

Find: R_2

✓ $R_{total} = V \div I = 9 \div 0.03 = 300\Omega$

✓ $R_2 = R_{total} - R_1 = 300 - 100 = 200\Omega$

✓ *Answer:* $R_2 = 200\Omega$

Q4 — Parallel Circuit — Find Total Resistance & Current

Two resistors are wired in parallel: $R_1 = 220\Omega$, $R_2 = 330\Omega$. Supply voltage is 9V. What is the total resistance and total current?

Given: $V = 9V$, $R_1 = 220\Omega$, $R_2 = 330\Omega$

Find: R_{total} , I_{total} , I_{R1} , I_{R2}

✓ $1/R_{total} = 1/220 + 1/330 = 3/660 + 2/660 = 5/660$

✓ $R_{total} = 660 \div 5 = 132\Omega$

✓ $I_{total} = V \div R_{total} = 9 \div 132 = 0.0682A \approx 68.2mA$

✓ $I_{R1} = 9 \div 220 = 40.9mA$

✓ $I_{R2} = 9 \div 330 = 27.3mA$

✓ *Check:* $40.9 + 27.3 = 68.2mA$ ✓

Q5 — Parallel Circuit — Find the Supply Voltage

Two resistors in parallel: $R_1 = 560\Omega$, $R_2 = 560\Omega$. The total current measured is 64.3mA. What is the supply voltage?

Given: $R_1 = R_2 = 560\Omega$, $I_{\text{total}} = 64.3\text{mA}$ (0.0643A)

Find: Supply Voltage (V)

✓ *Equal resistors in parallel: $R_{\text{total}} = 560 \div 2 = 280\Omega$*

✓ *$V = I \times R = 0.0643 \times 280 = 18\text{V}$*

✓ *Answer: $V = 18\text{V}$*

✓ *Each branch current = $18 \div 560 = 32.1\text{mA} \times 2 \text{ branches} = 64.3\text{mA}$ ✓*

Q6 — CURVEBALL: Series + Parallel Combined Circuit

This is the bonus challenge. A single resistor in series feeds into a parallel pair. Students must find the parallel equivalent first, add the series resistor, then solve for total current and individual branch currents. No one gets a pass on this one — if they can solve Q6 they understand the foundation of real circuit analysis.

Q6 — Series-Parallel — The Curveball

A 68Ω resistor (R1) is wired in series with two parallel resistors: $R2 = 100\Omega$ and $R3 = 100\Omega$. The supply voltage is 12V. Find: total resistance, total current, voltage across R1, voltage across the parallel pair, and current through each parallel branch.

Given: $V = 12V$, $R1 = 68\Omega$ (series), $R2 = 100\Omega$ (parallel), $R3 = 100\Omega$ (parallel)

Find: R_{total} , I_{total} , V_{R1} , $V_{parallel}$, I_{R2} , I_{R3}

- ✓ Step 1 — Parallel equivalent: $R_{parallel} = 100 \div 2 = 50\Omega$ (equal resistors)
- ✓ Step 2 — Total R: $R_{total} = R1 + R_{parallel} = 68 + 50 = 118\Omega$
- ✓ Step 3 — Total I: $I = V \div R_{total} = 12 \div 118 = 0.1017A \approx 101.7mA$
- ✓ Step 4 — V across R1: $V_{R1} = I \times R1 = 0.1017 \times 68 = 6.91V$
- ✓ Step 5 — V across parallel pair: $V_p = 12 - 6.91 = 5.09V$
- ✓ Step 6 — Branch currents: $I_{R2} = I_{R3} = 5.09 \div 100 = 50.9mA$ each
- ✓ Check: $50.9 + 50.9 = 101.7mA = I_{total}$ ✓

On-camera tip for Q6: Draw this on Xournal++ step by step. Start with the schematic — R1 in series, then the parallel pair. Solve the parallel section first, then treat it as a single resistor in series. Tell them: "This is how real circuits work — series and parallel combined. If you can solve this, you're thinking like a technician."

V8 — Closing Script & Giveaway Announcement

CAMERA: Canon wide bench shot. All gear visible. Look directly into camera for the close.

JOHN: Six questions. All Ohm's Law. All different values. Drop your answers in the comments — every question, every step of your math. No work shown, no entry.

JOHN: One subscriber who gets all six right and shows their work wins the FNIRSI DSO152 mini oscilloscope. This thing fits in your pocket and it reads waveforms. It's a real tool.

CAMERA: Hold up the FNIRSI DSO152 toward camera.

JOHN: I'll pick the winner and announce it in V9 — which is where we move on to Power. Watt's Law. Same style, same approach, and Q6 is going to start making a lot more sense once you understand how voltage and current relate to power dissipation.

JOHN: Subscribe. Do the math. Show your work. I'll see you in V9.